

Face Positioning

The main purpose of face positioning is to achieve face tracking as a functional advancement. The principle is to determine the distance and position information of the face relative to the camera, and to judge by calculating the center point coordinates of the face in the camera view, thereby achieving face positioning. In the experimental results, you can see that it will always follow the center point of the face, displaying the center point.

Source code path: /home/yahboom/dofbot_ws/src/dofbot_face_follow/scripts/Face_position.ipynb

Main Code

- Import header files

```
import cv2 as cv
import threading
from time import sleep
import ipywidgets as widgets
from IPython.display import display
from face_pose import face_follow
```

- Create instance, initialize parameters

```
# Create instance
follow = face_follow()
# Initialize mode
model = 'General'
```

- Main implementation function, print center coordinates.

```
def follow_function(self, img):

    img = cv.resize(img, (640, 480))
    # Copy the original image to avoid interference during processing
    img = img.copy()
    # Convert image to grayscale
    gray = cv.cvtColor(img, cv.COLOR_BGR2GRAY)
    # Face detection
    faces = self.faceDetect.detectMultiScale(gray, scaleFactor=1.3,
minNeighbors=5)
    if len(faces) != 0:
        face = self.face_filter(faces)
        # Face filtering
        (x, y, w, h) = face
        # Draw a rectangle on the original color map
        cv.rectangle(img, (x, y), (x + w, y + h), (0, 255, 0), 4)
        cv.putText(img, 'Person', (280, 30), cv.FONT_HERSHEY_SIMPLEX, 0.8,
(105, 105, 105), 2)
        point_x = x + w / 2
        point_y = y + h / 2

        print("x= ",point_x)
        print("y= ",point_y)
```

```
return img
```

- Main process

```
def camera():  
    global model  
    # Open camera  
    capture = cv.VideoCapture(0)  
    while capture.isOpened():  
        try:  
            _, img = capture.read()  
            img = cv.resize(img, (640, 480))  
            img = follow.follow_function(img)  
            if model == 'Exit':  
                cv.destroyAllWindows()  
                capture.release()  
                break  
            imgbox.value = cv.imencode('.jpg', img)[1].tobytes()  
        except KeyboardInterrupt: capture.release()
```

- Start

```
display(controls_box,output)  
threading.Thread(target=camera, ).start()
```

After the program starts, you can see the video display and face selection.

